

Part 2a: the p^+ integration by plus prescription

Yes — but you need it at *both* endpoints, and this row has no DGLAP in it

$$\eta = \frac{p^+}{k^+}, \quad \bar{\eta} = 1 - \eta, \quad A = (y - x)^2, \quad B = (y - z)^2, \quad \kappa = k \cdot (z - x), \quad \lambda = \frac{\Lambda}{k^+}, \quad h(\eta) = \frac{e^{-i\eta\kappa}}{\eta A + \bar{\eta} B}$$

Yes, and the answer is cleaner than Row IV's in one way and messier in another. The plus prescription is legitimate at both ends — the companion function $h(\eta)$ is smooth on the whole of $[0, 1]$, with $h(0) = \frac{1}{B}$ and $h(1) = \frac{e^{-i\kappa}}{A}$ both finite, and there is no pinch because $\eta A + \bar{\eta} B \geq \min(A, B) > 0$. But **this row has two endpoint poles, not one**: $\mathcal{T}_4$ goes as $-\frac{1}{\eta}$ at $p^+ \rightarrow 0$ and $\mathcal{T}_2$ as $-\frac{1}{\bar{\eta}}$ at $p^+ \rightarrow k^+$. So you get **two** δ -type terms and $2 \log \frac{k^+}{\Lambda}$ worth of rapidity logarithm. And do **not** change variables to $\xi = \frac{k^+}{k^+ + p^+}$: there is no fragmentation structure here at all.

1. Where the p^+ integrand is singular

The six tensors of eq. (3) are p^+ -independent. All the p^+ dependence is in three places: the six weights, the denominator $k^+[\eta A + \bar{\eta} B]$, and the phase $e^{-i\eta\kappa}$.

block	weight	$\eta \rightarrow 0$	$\eta \rightarrow 1$
$\mathcal{T}_1$	$\eta \bar{\eta}$	0	0
$\mathcal{T}_2$	$-\frac{\eta}{\bar{\eta}}$	0	$-\infty$
$\mathcal{T}_3$	$\bar{\eta}$	1	0
$\mathcal{T}_4$	$-\frac{\bar{\eta}}{\eta}$	$-\infty$	0
$\mathcal{T}_5$	η	0	1
$\mathcal{T}_6$	-1	-1	-1

Why two, when Row IV had one. This is a **loop** topology: $\mathfrak{C}_1$ splits the measured k^+ into p^+ and $k^+ - p^+$, and $\mathfrak{B}_2$ re-merges them. Both daughters can go soft, so both ends are singular and p^+ is bounded, $0 \leq p^+ \leq k^+$ — which is exactly why your limits are $\int_{\Lambda}^{k^+ - \Lambda}$ and not $\int_{\Lambda}^{\vee - k^+}$. In Row IV the measured gluon kept all of k^+ and the emitted one ran up to $\vee - k^+$, so there was one soft end only.

2. The prescription is legitimate at both ends

The test is the one from before: $\int \frac{d\eta}{\eta} [g(\eta) - g(0)]$ needs $g(0)$ finite. Here the residue of each pole is a (p^+ -independent) tensor times $h(\eta)$, and

$$\eta A + \bar{\eta} B \geq \min(A, B) > 0 \quad \text{for all } \eta \in [0, 1] \quad \implies \quad h \text{ smooth on } [0, 1],$$

$$h(0) = \frac{1}{(y - z)^2} \quad \text{finite}, \quad h(1) = \frac{e^{-ik \cdot (z - x)}}{(y - x)^2} \quad \text{finite}.$$

The denominator is *linear* in η and both A and B are positive, so there is no pinch anywhere in the interval — that is the whole content of the check, and it passes.

3. The result

$$\int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} (\dots) = \frac{k^+}{2\pi} \int_0^1 d\eta \left\{ - \left[\frac{1}{\eta} \right]_+ \bar{\eta} h(\eta) \mathcal{T}_4 - \left[\frac{1}{1 - \eta} \right]_+ \eta h(\eta) \mathcal{T}_2 + h(\eta) [\eta \bar{\eta} \mathcal{T}_1 + \bar{\eta} \mathcal{T}_3 + \eta \mathcal{T}_5 - \mathcal{T}_6] \right\} \\ - \frac{k^+}{2\pi} \log \frac{k^+}{\Lambda} \left[\frac{\mathcal{T}_4}{(y - z)^2} + \frac{e^{-ik \cdot (z - x)} \mathcal{T}_2}{(y - x)^2} \right]$$

Both δ terms carry the same logarithm, because $\int_{\lambda}^{1-\lambda} \frac{d\eta}{\eta} = \int_{\lambda}^{1-\lambda} \frac{d\eta}{1-\eta} = \log \frac{1-\lambda}{\lambda} \rightarrow \log \frac{k^+}{\Lambda}$. Checked against direct numerical integration:

Λ/k^+	direct	two-sided plus form	rel. difference
10^{-3}	$12.8377 + 4.2905i$	$12.8314 + 4.2895i$	4.8×10^{-4}
10^{-5}	$23.4840 + 7.3609i$	$23.4839 + 7.3609i$	2.6×10^{-6}
10^{-7}	$34.1341 + 10.4317i$	$34.1341 + 10.4317i$	1.8×10^{-8}
10^{-9}	$44.7843 + 13.5026i$	$44.7843 + 13.5026i$	3.2×10^{-10}

The residual is $O(\Lambda)$, the usual price of extending the regular integrals to the endpoints.

4. The two poles are mirror images — but you cannot fold them

Write the total phase as $\exp[-ik \cdot (w' - \bar{\eta}z - \eta x)]$ (which is what $e^{-ik \cdot (w' - z)} e^{-i\eta k \cdot (z - x)}$ is). Then the whole transverse-times-phase factor is invariant under $x \leftrightarrow z$ together with $p^+ \rightarrow k^+ - p^+$:

$$\max |F(x, y, z; p^+) - F(z, y, x; k^+ - p^+)| = 1.6 \times 10^{-13} \quad (200 \text{ configurations, } \max |F| = 586).$$

But the **colour** factors are not symmetric under $x \leftrightarrow z$ — not the original $f^{adc} f^{ebf} U^{db}(x) U^{ce}(z) + N_c U^{af}(y)$, and not $2a$'s $-f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)]$ or $2b$'s $N_c [U^{af}(y) - U^{af}(x)]$. So you may **not** use the symmetry to fold one pole onto the other and double it. Keep both terms explicitly.

5. The plus prescription does not spoil $2a$'s UV finiteness

Worth checking, because the δ coefficients are new objects. $u^2 \times \langle \delta\text{-coefficient} \times \text{colour} \rangle_u$ as $u = z - x \rightarrow 0$; a constant means log divergent:

$ z - x $	$2a, \eta \rightarrow 0$	$2a, \eta \rightarrow 1$	$2b, \eta \rightarrow 0$	$2b, \eta \rightarrow 1$
10^{-2}	1.16×10^{-5}	1.37×10^{-6}	0.02131563	0.02131565
10^{-3}	1.16×10^{-7}	1.37×10^{-8}	0.02131583	0.02131583
10^{-4}	1.16×10^{-9}	1.37×10^{-10}	0.02131583	0.02131583

In $2a$ both δ coefficients are UV finite — $\mathcal{C}_{2a} = O(|z - x|)$ still does its job, so ordering does not matter and you can do the p^+ integral first. In $2b$ they are not, and the two divergences **multiply**: you will get $\log_{\text{UV}} \times \log \frac{k^+}{\Lambda}$. That is exactly consistent with $C_{\text{UV}}(\eta) = \eta \bar{\eta} + \frac{\eta}{\bar{\eta}} + \frac{\bar{\eta}}{\eta}$ having unit-residue poles at *both* endpoints — which is why the two $2b$ columns above agree to seven digits. **For $2b$, regulate the transverse UV first, then do p^+ .**

6. One thing not to do: no ξ change of variable here

In Row IV the substitution $\xi = \frac{k^+}{k^+ + p^+}$ produced the fragmentation measure $\frac{d\xi}{\xi}$, a hard factor at $\frac{k}{\xi}$, and $\mathcal{P}_{gg}(\xi)$. **None of that happens here.** In this row $p^+ \leq k^+$, so ξ would only cover $[\frac{1}{2}, 1]$, and the measured momentum never leaves k : the phase is $\exp[-ik \cdot (w' - \bar{\eta}z - \eta x)]$, with no $\frac{k}{\xi}$ anywhere. There is no $\frac{d\xi}{\xi}$ and no convolution with a hard factor at a shifted momentum, so **this row carries no DGLAP**. It is a virtual correction: everything it produces is $\delta(1 - \xi)$ -like in the factorisation sense. $\mathcal{P}_{gg}$ did appear in it, but as the *transverse UV residue* of part $2b$, not as a splitting convolution in p^+ .

So the natural variable is $\eta = \frac{p^+}{k^+} \in [0, 1]$, the natural form is the two-sided plus prescription above, and the output is $2 \log \frac{k^+}{\Lambda}$ times two z - and x -dependent coefficients plus a finite remainder. The $\log \frac{k^+}{\Lambda}$ terms are the rapidity divergence of the loop — they belong with the JIMWLK evolution of the target, alongside Row IV's, and their Λ must cancel there.

Checks: `src/rowB2_pplus.py` — §1 the weights and their endpoint limits in sympy; §2 the no-pinch argument; §3 the two-sided plus form against direct mpmath quadrature at four values of Λ ; §4 the $(x \leftrightarrow z, p^+ \leftrightarrow k^+ - p^+)$ symmetry over 200 configurations; §5 the UV scan of both δ coefficients for $2a$ and $2b$. Output in `RESULTS_rowB2_pplus.txt`.